

Wizard Chess Presentation

Embedded Systems, Cyber-
Physical Systems and Robotics
Project

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Wizard Chess

OUR TEAM

Habiba



Alexandra



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Eyas



Tan



Yahya





Wizard Chess Introduction

Wizard Chess is an autonomous chessboard inspired by the magical chess scenes of Harry Potter. The system brings the concept to life by allowing players to make moves through voice or digital commands while the physical chess pieces move automatically across the board.

Use Cases

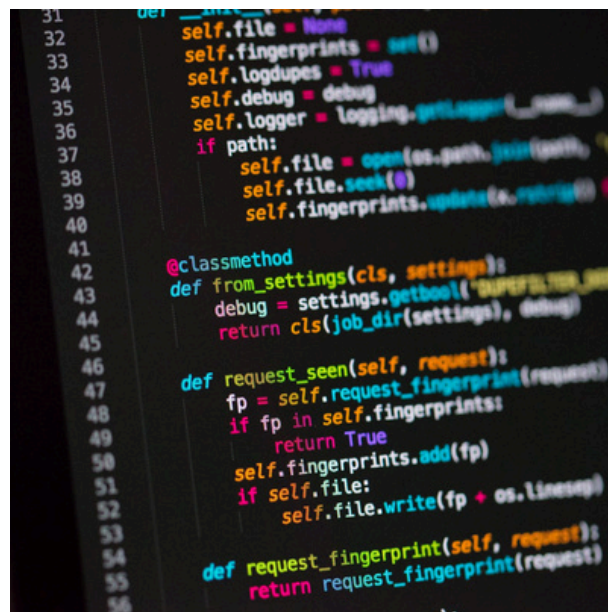
01 Interactive chess

02 Blind chess

03 Chess education

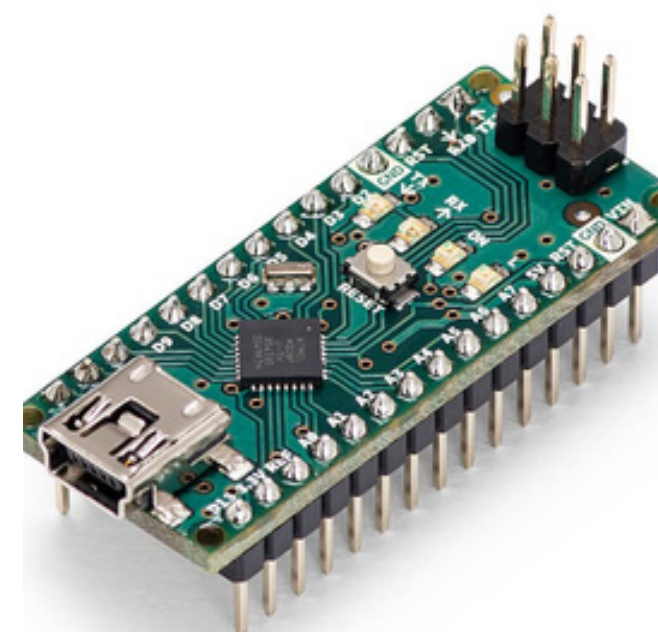


Components



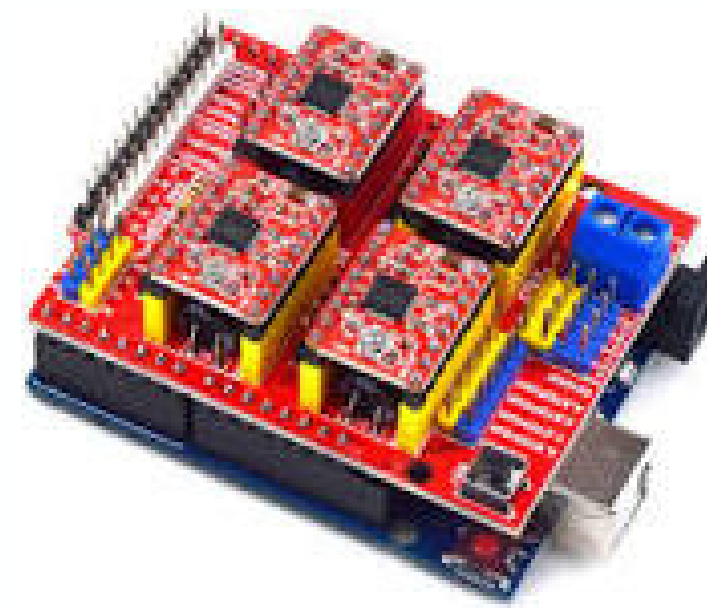
Python Application

Main controller connecting user input, Stockfish, and hardware.



Arduino Uno + GRBL

Executes movement commands received from Python.



CNC Shield + DRV8825

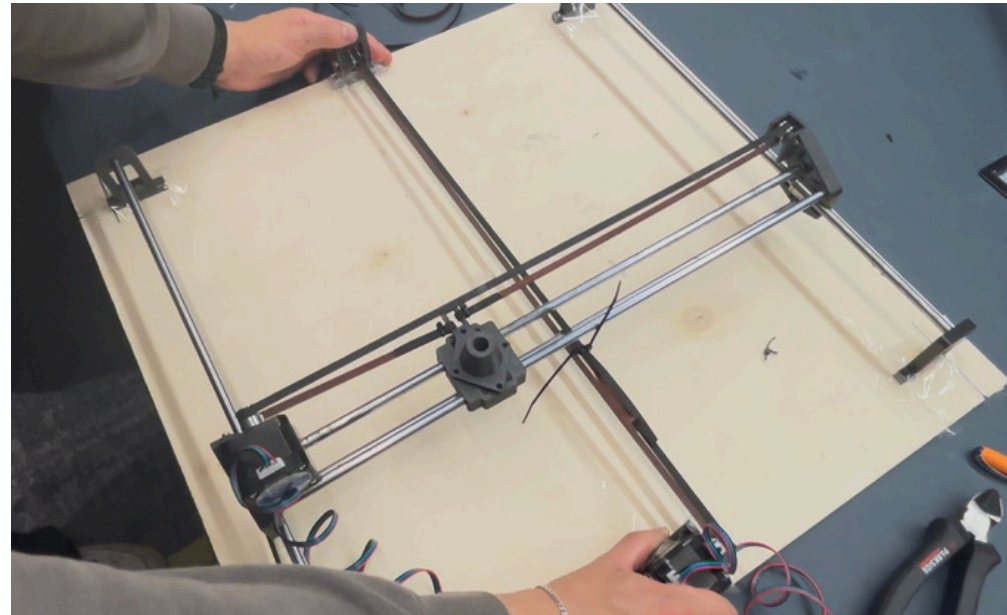
Controls and drives the stepper motors.



NEMA 17 Motors

Provide precise X–Y movement beneath the board.

Components



XY Cartesian System

One motor controls movement along the X-axis, while the other controls the Y-axis



5V Electromagnet

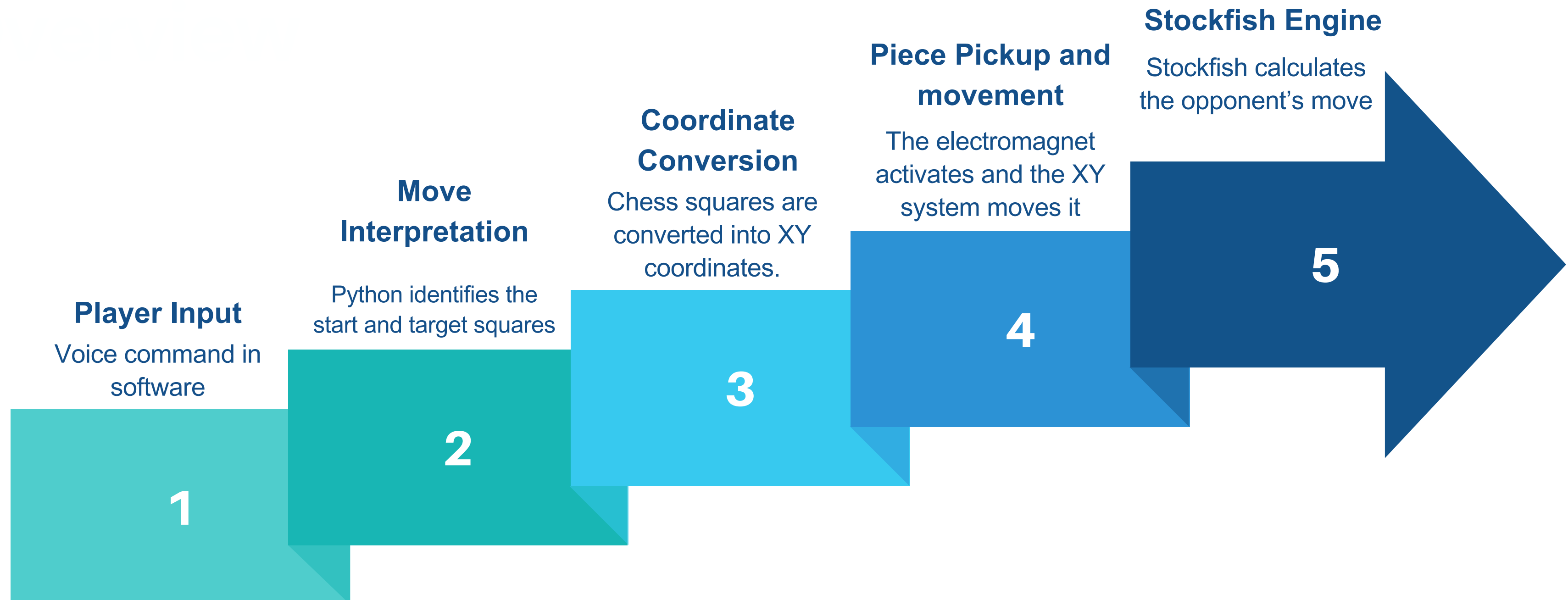
Picks up, transports, and releases chess pieces.



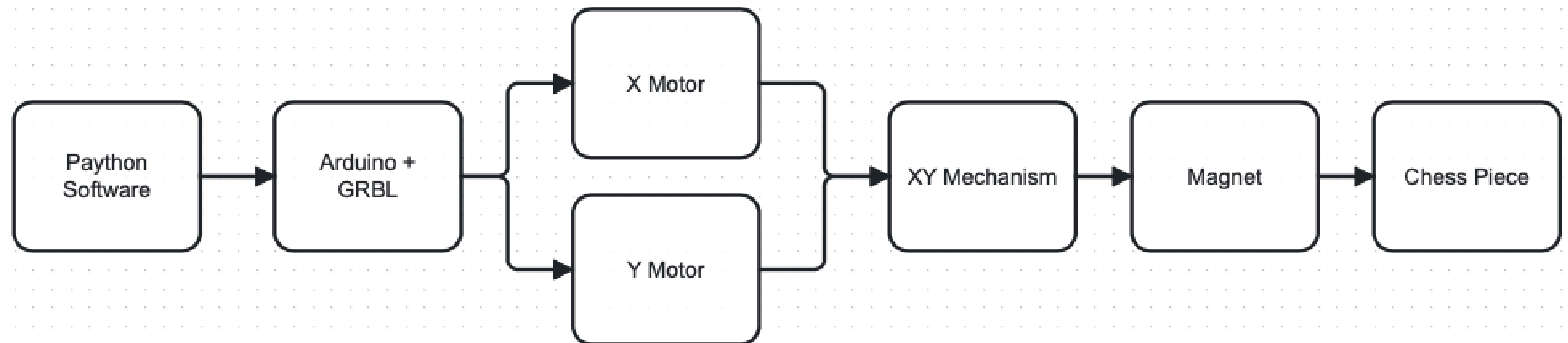
Custom board & pieces

Laser-cut board with 3D-printed themed chess pieces.

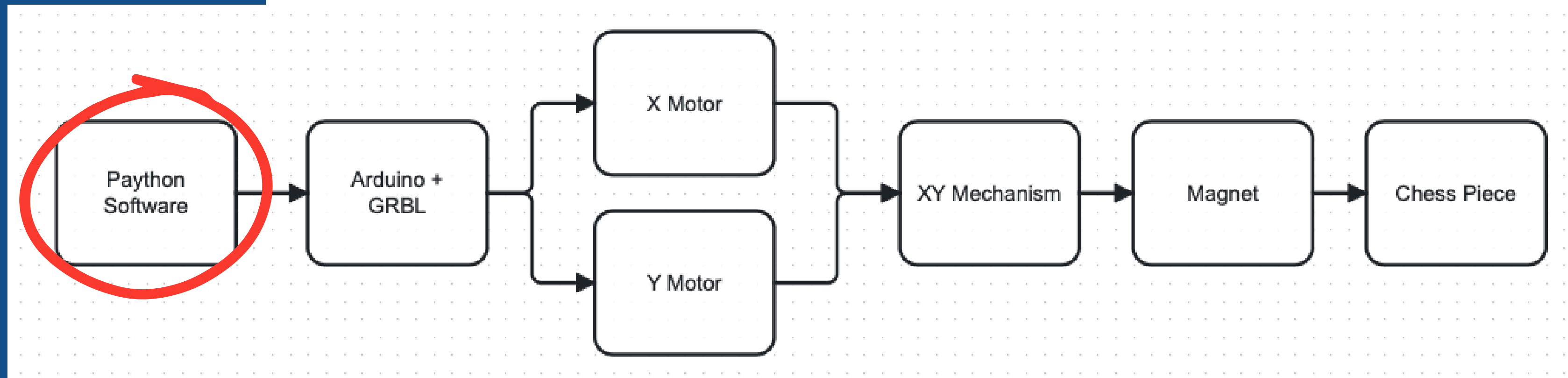
Workflow



System Overview

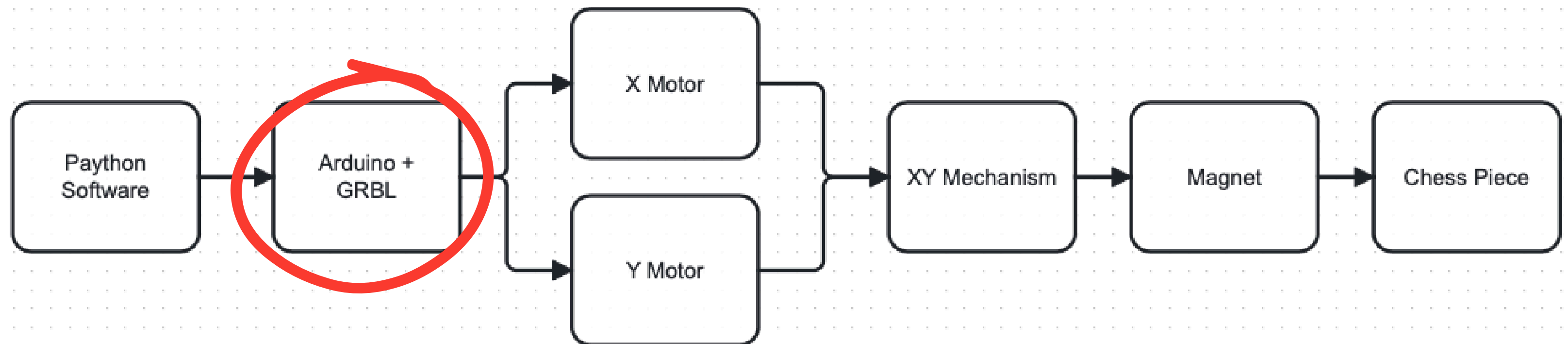


System Overview



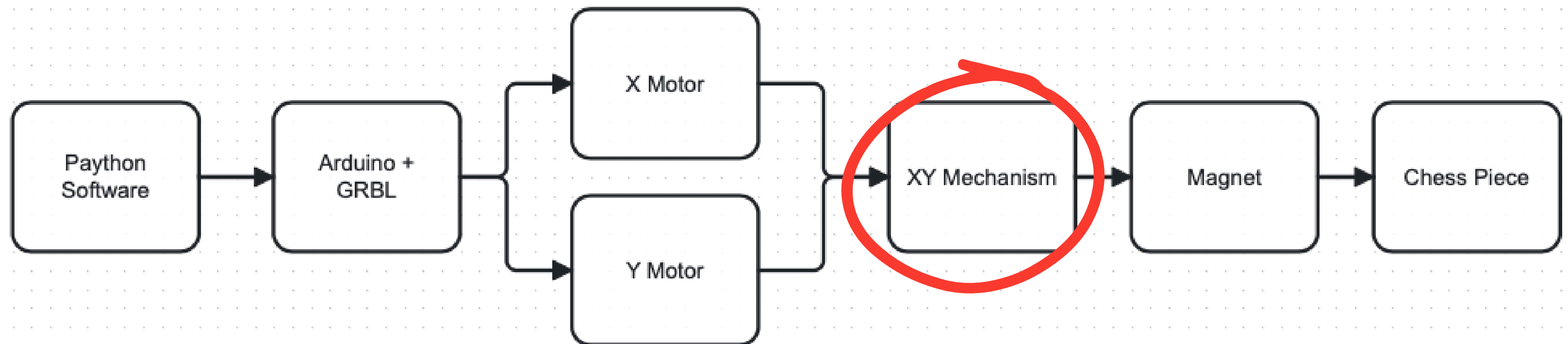
- Receives voice /move command
- Converts chess move into XY coordinates
- eg G0 X70 Y100

System Overview



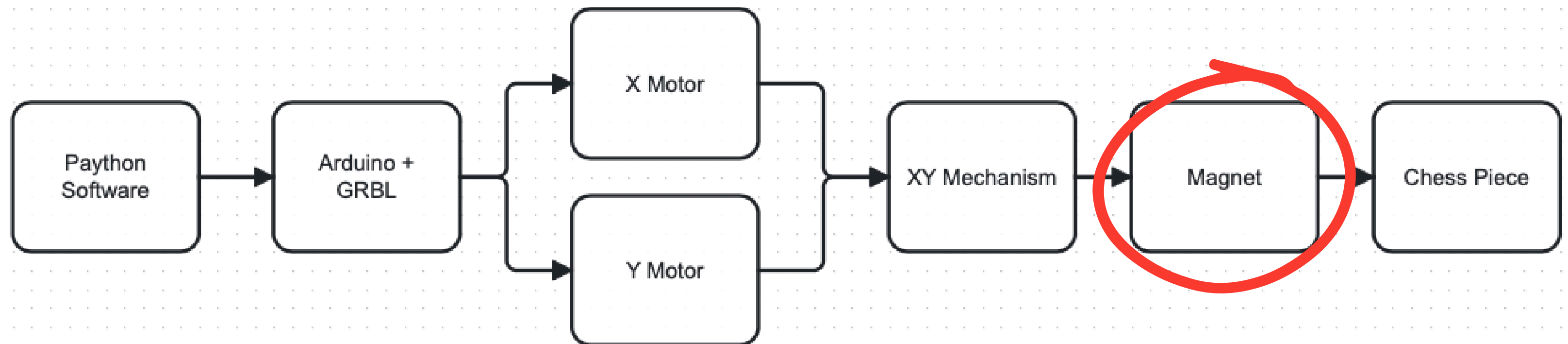
- Receives movement commands via serial
- Generates step & direction signals

System Overview



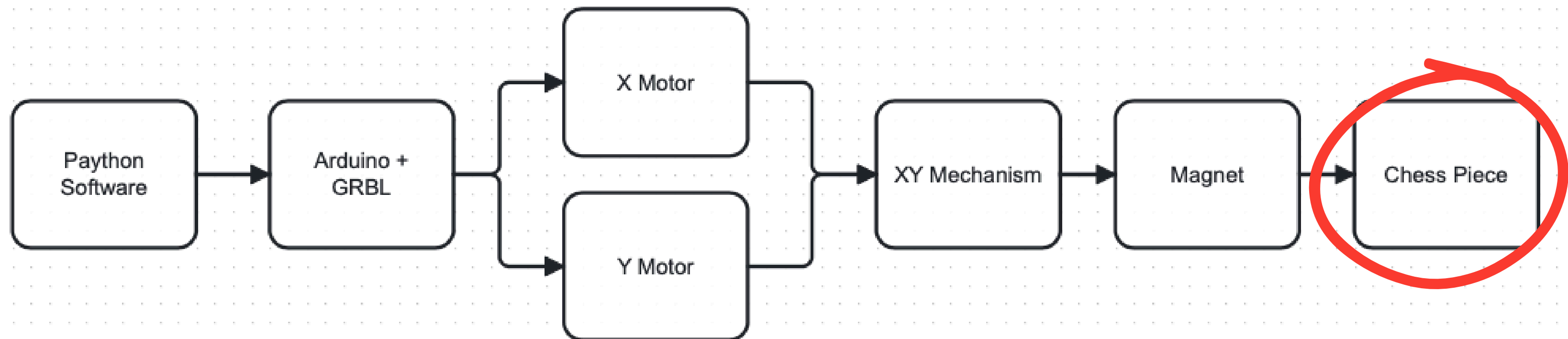
- Converts motor rotation into 2D Cartesian movement
- Positions the carriage under the target square

System Overview



- Activates when under piece (M3 S1000)
- Deactivates when piece at target (M5)

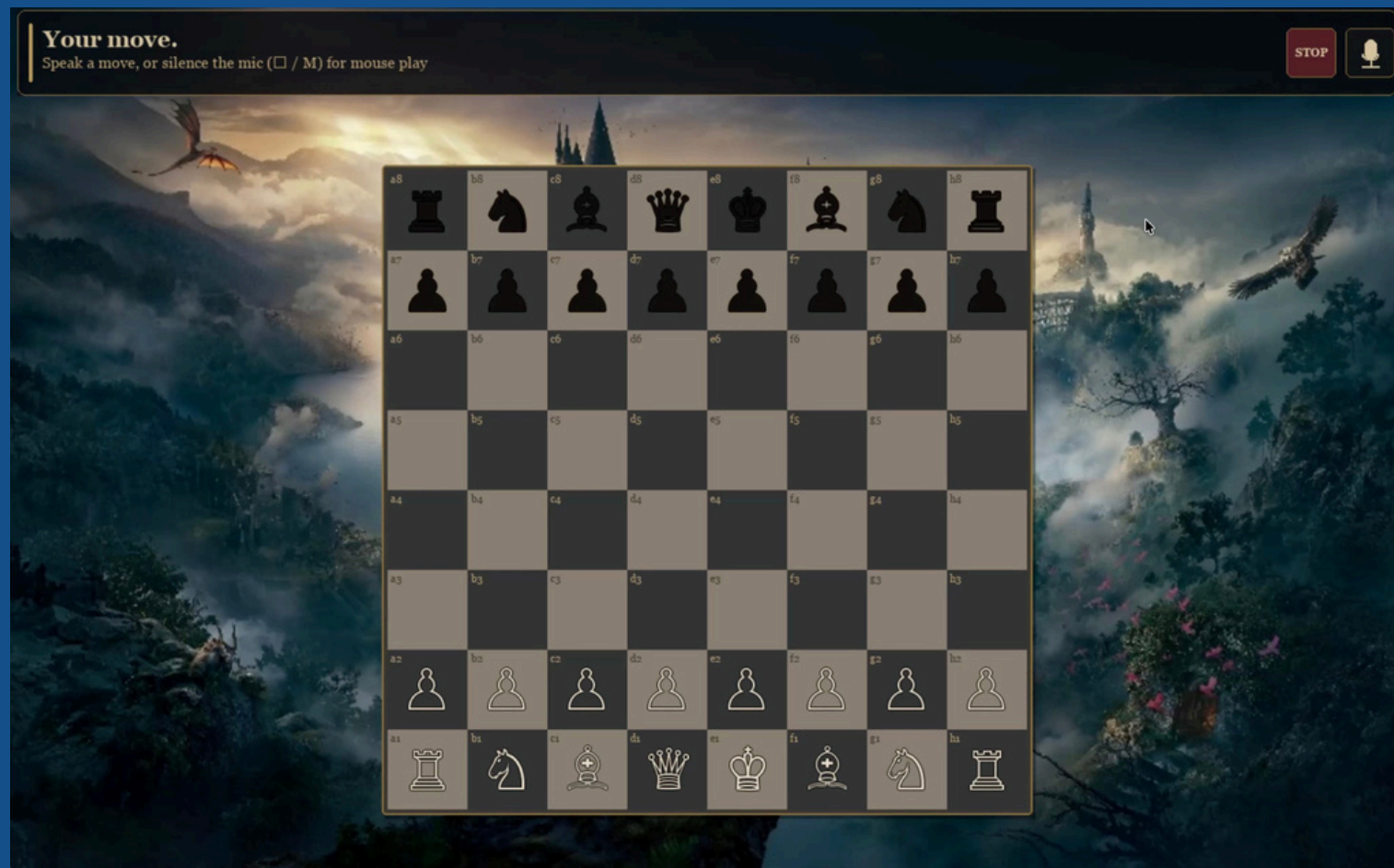
System Overview



- Contains a magnet embedded underneath so magnet can move it

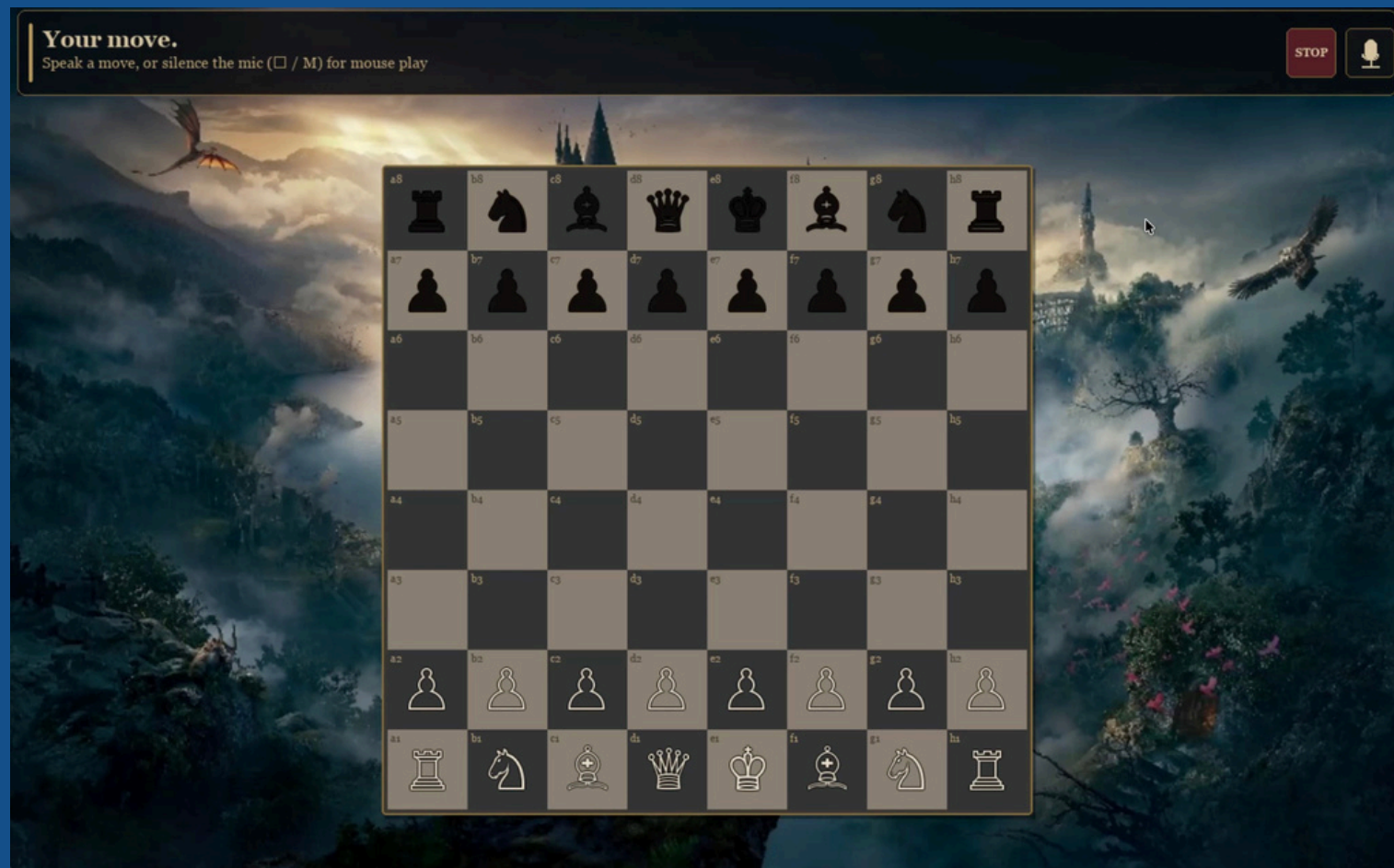
Implementation

Python Software



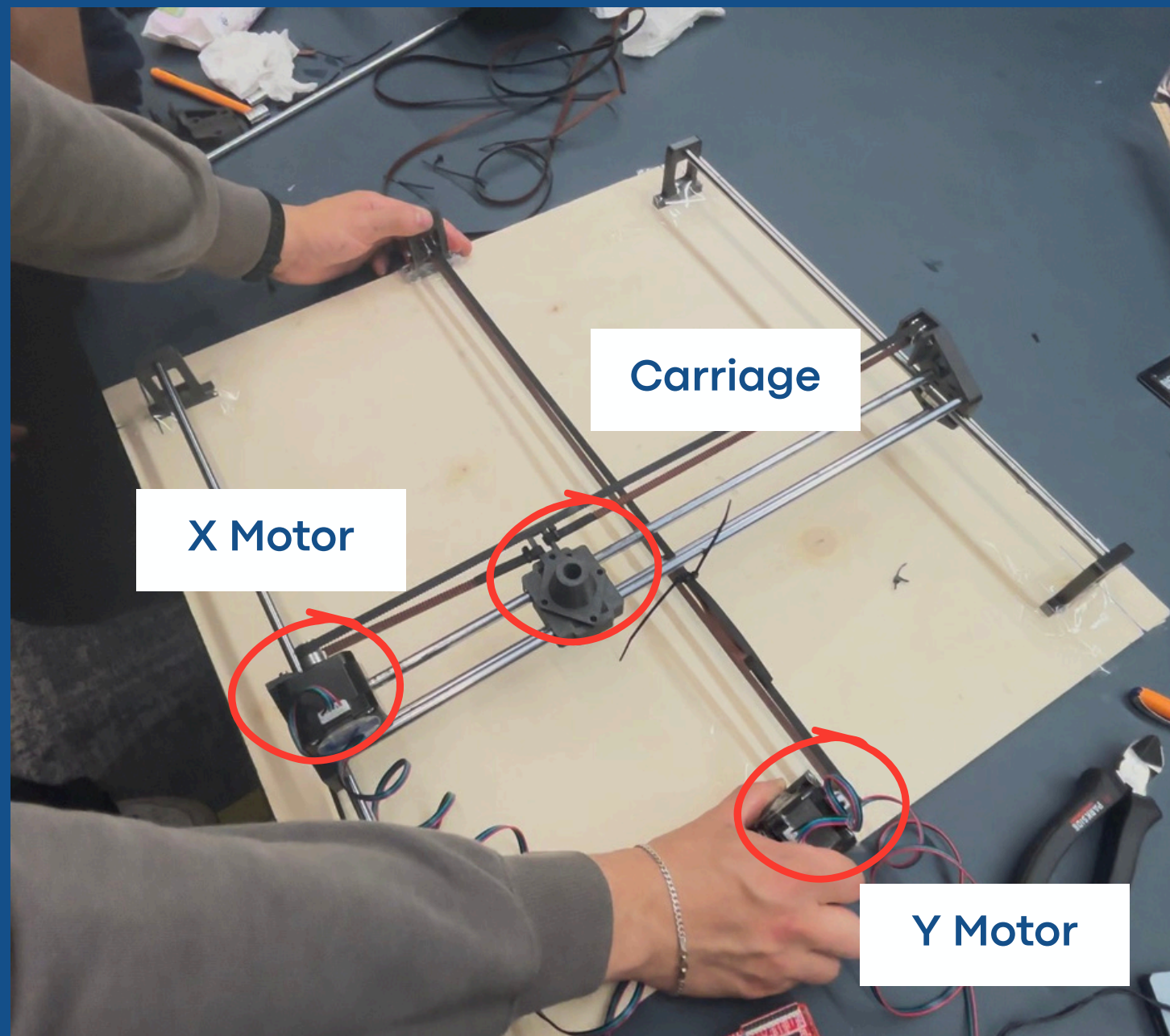
- FastWhisper converts voice commands into chess moves.
- Stockfish engine validates moves and provides game logic.
- Converts chess squares (e.g., E2 → E4) into physical XY coordinates.
- Generates movement paths for normal moves and captures.
- Sends G-code commands to GRBL via serial communication.

Special Movement Handling



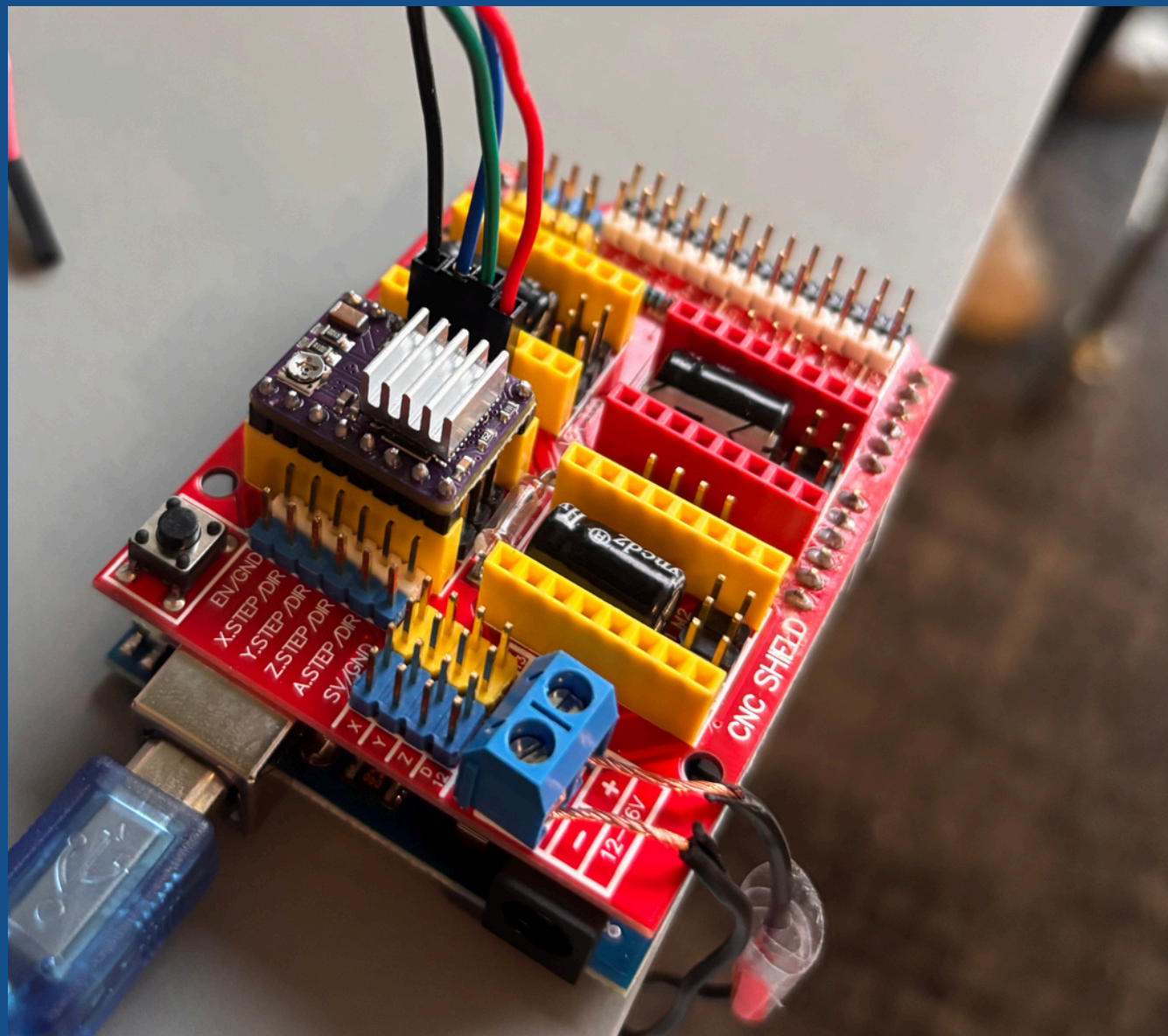
- Knight moves: Path planning finds a short collision-free route between surrounding pieces.
- Captures: Moves the captured piece off the board first, then moves the attacking piece to its position.
- Path planning: Generates intermediate XY coordinates to avoid collisions with other pieces.

Mechanical System



- 3D printed mounts support motors and carriage.
- X and Y motors control movement along each axis.
- Carriage holds the electromagnet at a fixed height.
- Bottom-left position defined as origin (X0 Y0).
- UGS used for movement testing and calibration.

Embedded System



- Arduino Uno + CNC Shield V3 form the embedded motion controller.
- GRBL firmware interprets incoming G-code commands.
- DRV8825 drivers control the NEMA 17 stepper motors.
- Generates precise step/direction signals for XY movement.
- Interfaces the software commands with the physical XY mechanism.

Challenges and Future Improvements

Challenges

- **XY Calibration:** Fine-tuned motor movement so the carriage stops precisely at the center of each chess square.
- **Board Alignment:** Positioned the board relative to the carriage origin (0,0) to ensure software coordinates matched physical squares.
- **Electromagnet Calibration:** Adjusted magnet height and power to reliably move pieces of different weights while allowing clean release.
- **Piece Friction:** Minimized friction at the base of the pieces to ensure smooth movement and prevent positioning errors.

Future Improvements

- **Two-Player Mode:** Enable two players to play against each other while the system automatically executes their moves.
- **Computer Vision:** Use a camera to detect physical piece movements and board state as an alternative input method.
- **Smart Board Detection:** Integrate a sensor/magnet PCB beneath each square to automatically detect when and where pieces are moved.

Wizard Chess

Any Questions?

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Thank you!

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